

Technical note: Fisher-matrix estimates from ASTRA environment-split clustering

ASTRA clustering project

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Abstract

We present the setup and first results of a Fisher-matrix forecast based on the two-point clustering of an HOD galaxy sample split by cosmic-web environment with the ASTRA classification. Derivatives with respect to the cosmological parameters come from central differences over the AbacusSummit linear derivative grid; the covariance comes from 64 subboxes of the fiducial simulation. With the ω_b pair complete, we compare the constraining power of the individual auto- and cross-correlations. The cross-correlation of the full sample with the overdense environment quantiles carries slightly more information than the full-sample auto-correlation itself. All absolute uncertainties are still optimistic because the available HOD catalogs are not matched across cosmologies.

1 Method

For a single parameter θ , the Fisher information of a data vector $\boldsymbol{\xi}$ with covariance $\mathbf{C}$ is

$$F = \frac{\partial \boldsymbol{\xi}^\top}{\partial \theta} \mathbf{C}^{-1} \frac{\partial \boldsymbol{\xi}}{\partial \theta}, \quad \sigma(\theta) = F^{-1/2} \quad (1)$$

[1]. Our data vectors are built from the redshift-space 2PCF multipoles ($\ell = 0, 2$; 15 bins over $0 < s < 150 h^{-1} \text{ Mpc}$) of the full LRG-like sample, of the four ASTRA environment quantiles (Q1 = most underdense, Q4 = most overdense), and of their cross-correlations with the full sample. The measurement pipeline (Landy–Szalay estimator over a $4 \times 4 \times 4$ grid of $500 h^{-1} \text{ Mpc}$ subboxes) is described in the companion zero-crossing note.

The derivative for each parameter comes from the simulation pair that brackets the fiducial cosmology. We difference the *same* subbox i in the two cosmologies before averaging,

$$\frac{\partial \xi_i}{\partial \theta} = \frac{\xi_i(\theta + \Delta\theta) - \xi_i(\theta - \Delta\theta)}{2 \Delta\theta}. \quad (2)$$

Because all derivative-grid simulations share the initial-condition phases of the fiducial box, and our pipelines also share all random seeds, most of the cosmic variance cancels in Eq. (2); the paired estimate is about twice as precise as differencing independent volumes. The mean over the 64 subboxes gives the derivative and its scatter gives the derivative uncertainty.

The covariance $\mathbf{C}$ is estimated from the 64 fiducial subboxes. An inverse covariance built from n samples and p bins is biased, so we apply the standard correction $(n - p - 2)/(n - 1)$ [3]. Noise in the derivative adds spurious Fisher information with expectation $\text{tr}(\mathbf{C}^{-1} \boldsymbol{\Sigma}_d)$, where $\boldsymbol{\Sigma}_d$ is the covariance of the derivative estimate; we subtract this term and quote it as the “noise share” of each data vector.

Figure 1 shows the covariance matrices that enter Eq. (1), in their normalised form $R_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$ (the raw covariance spans too many orders of magnitude to display). The single-statistic matrices show the expected structure: neighbouring separation bins are strongly correlated, and the correlation length grows toward large s , where the same long-wavelength modes dominate every bin. The concatenated vector makes the cross-blocks visible. The full auto and the full $\times$ data Q4 blocks are almost perfectly correlated with each other — this is the redundancy discussed in Section 4. In contrast, the full $\times$ random Q1 block is strongly *anti*-correlated with the other two: when a subbox scatters high in clustering amplitude, its underdense-environment cross scatters low. This anti-correlation is the only genuinely complementary information in the concatenation.

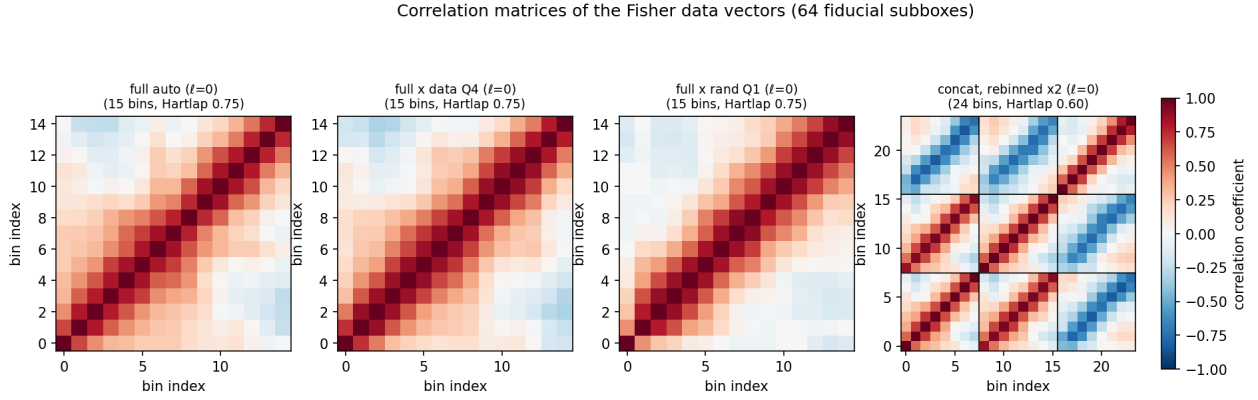


Figure 1: Correlation matrices of the data vectors used in the Fisher comparison, estimated from the 64 fiducial subboxes. Black lines in the right panel separate the three concatenated pieces (full auto, full $\times$ data Q4, full $\times$ random Q1; 8 rebinned bins each). Panel titles give the number of bins and the corresponding $(n - p - 2)/(n - 1)$ correction factor.

From σ to the plotted curves. The curves we show later are not measured posteriors; they are the likelihood implied by the Fisher formalism. In one dimension the Fisher approach expands the log-likelihood to second order around its maximum, which is placed at the fiducial value θ_{fid} . The quadratic term is the Fisher information F , and every higher-order term is dropped, so the likelihood is *by construction* a Gaussian,

$$\mathcal{L}(\theta) \propto \exp\left[-\frac{1}{2}F(\theta - \theta_{\text{fid}})^2\right] = \exp\left[-\frac{(\theta - \theta_{\text{fid}})^2}{2\sigma^2}\right], \quad \sigma = F^{-1/2}. \quad (3)$$

This is exact when the data are Gaussian-distributed and the model $\xi(\theta)$ is linear in θ over the range of interest; otherwise it is the leading approximation. We plot Eq. (3) normalised to unit peak, so only the width carries information. When the derivative is taken with respect to a logarithm, such as $\ln\omega_b$, the width in the physical parameter follows from the same expansion: $\sigma(\omega_b) = \omega_{b,\text{fid}} \sigma(\ln\omega_b)$.

2 Simulations

Table 1 lists the nine simulations of the design: the fiducial cosmology and the four $\pm$ pairs of the AbacusSummit linear derivative grid [2], all at phase **ph000**, $z = 0.5$. Each is populated with an LRG-like HOD catalog ($\sim 4 \times 10^6$ galaxies) from the EMC emulator set. In that set, every cosmology carries its own random draws from the HOD prior, so no identical HOD exists across

cosmologies. The catalogs in Table 1 were selected to minimise the HOD-parameter mismatch inside each $\pm$ pair around a common mid-prior anchor point (pair distances 0.14–0.16 of the prior range, against 0.40 for random draws). Galaxy number density is matched by construction.

Cosmology	Variation	HOD catalog	Status
c000	fiducial (Planck 2018 Λ CDM)	hod484	complete
c100 / c101	$\pm 2\%$ in $\ln \omega_b$	hod179 / hod152	complete
c102 / c103	$\pm 3.3\%$ in $\ln \omega_{cdm}$	hod556 / hod861	queued
c104 / c105	± 0.01 in n_s	hod498 / hod589	queued
c112 / c113	$\pm 2\%$ in σ_8	hod507 / hod483	queued

Table 1: Simulations of the 4-parameter Λ CDM Fisher design. All are AbacusSummit base boxes ($2000 h^{-1}$ Mpc, phase ph000). The grid holds the CMB acoustic scale θ_* fixed (h is retuned) and, except for the σ_8 pair, retunes A_s so that σ_8 stays constant. Status at the time of writing.

The residual HOD mismatch matters. The measured c100–c101 difference is a scale-dependent -6% to $+9\%$ tilt of ξ_0 , while the physical ω_b signal at fixed σ_8 and θ_* is only $\sim 1\%$. The derivative is therefore still dominated by the different galaxy–halo connections of the two catalogs, and it is not a flat offset, so a simple amplitude nuisance cannot absorb it completely. All absolute uncertainties in this note are too small for this reason; we read them as a *relative* comparison between data vectors. A planned correction is to measure the HOD response $\partial\xi/\partial\theta_{\text{HOD}}$ from additional existing fiducial catalogs and subtract it.

3 Derivatives

Figure 2 shows the central-difference derivative of every statistic with respect to $\ln \omega_b$, from Eq. (2) applied to the 64 paired subboxes. The derivative of the full-sample auto-correlation is detected with a signal-to-noise ratio of up to ~ 19 per bin at small separations; at $s \gtrsim 80 h^{-1}$ Mpc several bins are consistent with zero. The quantile statistics inherit the same tilt with larger noise. These are the inputs to Eq. (1).

4 Results

Table 2 lists the one-parameter uncertainties for the six best-measured data vectors, and Figure 3 shows a selection as Gaussian likelihoods around the fiducial value.

Three lessons emerge. First, the cross-correlations of the full sample with the overdense quantiles (Q3, Q4) carry slightly more information than the full-sample auto-correlation itself: the cross keeps the low shot noise of the full sample while the environment selection sharpens the response. Second, the quadrupole adds almost nothing for ω_b : doubling the number of bins costs more through the $(n - p - 2)/(n - 1)$ factor than the new bins contribute. Third, concatenating vectors that share the full-sample leg (full auto $\oplus$ full $\times$ data Q4 $\oplus$ full $\times$ random Q1) ties with the full auto alone, even after compressing each piece from 15 to 8 bins to soften the covariance penalty: the pieces are too correlated. Real gains must come from independent tracers in the data vector or from more covariance realisations.

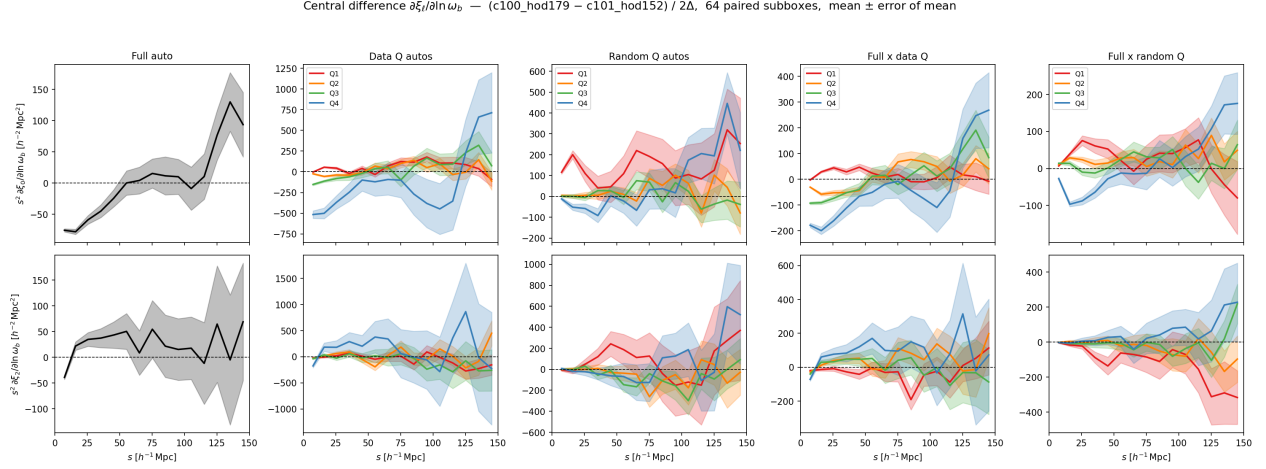


Figure 2: Central-difference derivatives $\partial\xi_\ell/\partial\ln\omega_b$ for all measured statistics (columns: full auto, data quantile autos, random quantile autos, full $\times$ data quantiles, full $\times$ random quantiles; rows: monopole and quadrupole). Lines are the mean over 64 paired subboxes; bands are the error of the mean.

Data vector ($\ell = 0, 2$)	$\sigma(\ln\omega_b)$ per subbox	noise share
Full $\times$ data Q3	0.0092	3%
Full auto	0.0102	3%
Full $\times$ data Q4	0.0107	4%
Random Q1 auto	0.0115	4%
Data Q3 auto	0.0123	6%
Data Q4 auto	0.0151	7%

Table 2: One-parameter Fisher uncertainties for $\ln\omega_b$ from a single $500 h^{-1}$ Mpc subbox volume (divide by 8 for the full box), with the Hartlap correction applied and the derivative-noise information subtracted. “Noise share” is the subtracted fraction. Statistics not listed are noise-dominated at the current derivative precision.

5 Next steps

The remaining pairs of Table 1 complete the 4-parameter design; the same scripts process them automatically. The σ_8 pair is the most important test: its physical signal ($\sim 8\%$ in amplitude across the pair) should dominate both the noise and the residual HOD contamination, unlike ω_b . The two structural improvements identified here are an HOD-response correction built from additional fiducial catalogs, and a larger set of covariance realisations so that longer concatenated data vectors become usable.

References

- [1] M. Tegmark, A. N. Taylor, A. F. Heavens, *Karhunen-Loève eigenvalue problems in cosmology: how should we tackle large data sets?*, ApJ 480, 22 (1997).

Fisher forecasts for ω_b per data vector (64-subbox covariance, Hartlap-corrected, derivative-noise bias subtracted)

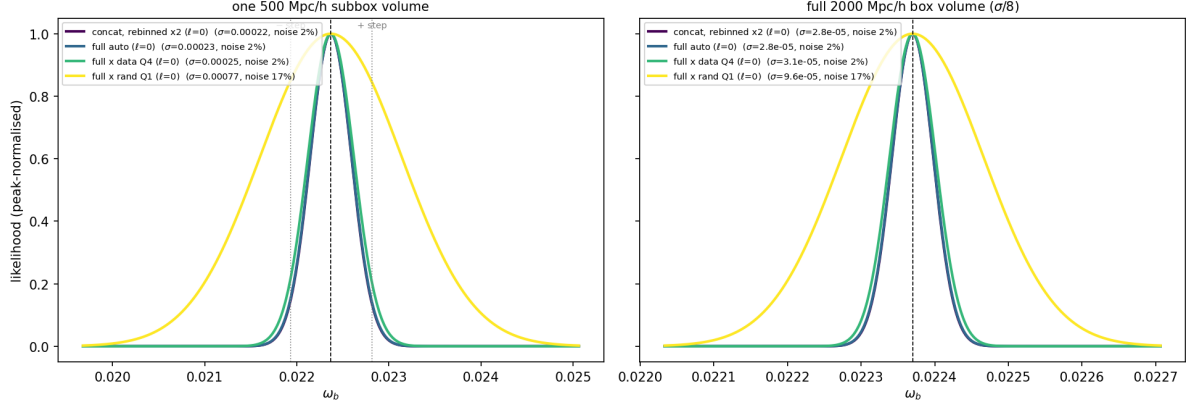


Figure 3: Fisher uncertainties for ω_b shown as Gaussian likelihoods around the fiducial value, for one subbox volume (left) and the full box (right). The concatenated vector ties with the full-sample auto-correlation; the underdense-random cross is several times weaker. All widths are optimistic because of the HOD mismatch discussed in Section 2.

- [2] N. A. Maksimova et al., *AbacusSummit: a massive set of high-accuracy, high-resolution N-body simulations*, MNRAS 508, 4017 (2021).
- [3] J. Hartlap, P. Simon, P. Schneider, *Why your model parameter confidences may be too optimistic. Unbiased estimation of the inverse covariance matrix*, A&A 464, 399 (2007).